ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

COMPUTER PROGRAMMING

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 4. Section A: objective questions (30) · Section B: object-oriented programming, string handling, searching and sorting, text files and exception handling (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Code in this paper is written in Python. You may write your answers in Python, C#, Java or pseudocode – state which language you use. Minor syntax errors will not be penalised if the logic is correct.
5. Write neatly and legibly. Indent your code clearly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A blueprint from which objects are created is a …

A method.
B module.
C variable.
D class.

1.1.2 Hiding the data of an object and allowing access only through methods is …

A inheritance.
B encapsulation.
C polymorphism.
D iteration.

1.1.3 A special method that runs automatically when an object is created is the …

A destructor.
B accessor.
C constructor.
D loop.

1.1.4 A Car class that gets all the attributes of a Vehicle class is an example of …

A inheritance.
B overloading.
C encapsulation.
D recursion.

1.1.5 A binary search …

A checks every element from the first to the last.
B only works on a sorted list.
C works best on unsorted lists.
D can only search numbers.

1.1.6 The value of "Polokwane"[2:5] in Python is …

A "olo".
B "loka".
C "Pol".
D "lok".

1.1.7 Opening a text file in mode "a" will …

A add new data to the end of the file.
B delete the file.
C open it for reading only.
D create a binary file.

1.1.8 The block of code that handles an error so that the program does not crash is the …

A else block.
B loop body.
C except (catch) block.
D return statement.

1.1.9 How many comparisons does a linear search need, in the worst case, for a list of 50 items?

A 25
B 6
C 50
D 1

1.1.10 A method that returns the value of a private attribute is called a/an …

A getter (accessor).
B setter (mutator).
C constructor.
D parameter.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Many objects can be created from one class.

1.2.2 Polymorphism allows the same method name to behave differently in different classes.

1.2.3 A bubble sort repeatedly swaps adjacent elements that are in the wrong order.

1.2.4 Opening an existing file in mode "w" keeps the old contents.

1.2.5 "abc".upper() returns "ABC".

1.2.6 Binary search is generally slower than linear search on large sorted lists.

1.2.7 A file should be closed after it has been used.

1.2.8 Attributes describe what an object can do, and methods describe what it has.

1.2.9 A ValueError occurs when int("ten") is executed in Python.

1.2.10 The str method of a class can be used to return a readable description of an object.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	An instance of a class	A	strip
1.3.2	Data stored inside an object	B	setter (mutator)
1.3.3	A function that belongs to a class	C	UML class diagram
1.3.4	The class that another class inherits from	D	FileNotFoundError
1.3.5	A method that changes the value of a private attribute	E	parent (super) class
1.3.6	Splitting a line of text into parts at each comma	F	recursion
1.3.7	A function that calls itself	G	method
1.3.8	Error raised when a file that does not exist is opened for reading	H	object
1.3.9	Visual diagram showing a class's name, attributes and methods	I	split
1.3.10	Removing spaces and line breaks from the ends of a string	J	attribute

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: OBJECT-ORIENTED PROGRAMMING

2.1 A college wants a program to manage student accounts. Study the UML class diagram and answer the questions.

```
+----------------------------------------+
|                Student                 |
+----------------------------------------+
| - studentNo : String                   |
| - name : String                        |
| - feesOwed : Real                      |
+----------------------------------------+
| + Student(studentNo, name, feesOwed)   |
| + getName() : String                   |
| + makePayment(amount : Real) : void    |
| + isPaidUp() : Boolean                 |
| + toString() : String                  |
+----------------------------------------+
```

2.1.1 What do the symbols – and + mean in the diagram? (2)

2.1.2 Write the code for the class with the constructor and the getName method. (5)

2.1.3 Write the makePayment method. It must reduce feesOwed by the amount, but only if the amount is positive and not more than the fees owed; otherwise it must display "Invalid payment". (5)

2.1.4 Write the isPaidUp method, which returns True when no fees are owed. (2)

2.1.5 Write the toString method, which returns a string such as "202401234 Thabo Nkosi owes R1500.00". (3)

2.1.6 Write code in the main program to create a Student object for student 202401234, Thabo Nkosi, who owes R4 500, make a payment of R3 000 and display the object. (4)

2.2 A class Bursar inherits from Student and adds an attribute bursaryName. Explain TWO advantages of using inheritance here. (2)

2.3 Explain why the attribute feesOwed is made private. (2)

[25]

QUESTION 3: STRINGS, SEARCHING AND SORTING

3.1 Write a function countVowels(word) that returns the number of vowels (a, e, i, o, u – upper or lower case) in a string. (5)

3.2 The list below must be sorted in ascending order with a bubble sort.

```
nums = [27, 12, 45, 8, 19]
```

3.2.1 Show the list after EACH pass of the bubble sort. (4)

3.2.2 Write the code for the bubble sort. (6)

3.3 Use the sorted list from QUESTION 3.2 to explain, step by step, how a binary search would find the value 27. (4)

3.4 Give ONE advantage and ONE disadvantage of a binary search compared with a linear search. (2)

[21]

QUESTION 4: TEXT FILES AND EXCEPTION HANDLING

4.1 The text file *results.txt* contains one student per line, in the format studentNo,name,mark. For example:

```
202401234,Thabo Nkosi,67
202401235,Ayanda Dube,82
202401236,Pieter Smit,45
```

4.1.1 Write code that reads the file, and displays the name of every student who obtained 50 or more, and finally the class average. (8)

4.1.2 Add exception handling to your code so that a friendly message is shown if the file does not exist, or if a mark is not a valid number. (4)

4.2 Write code that appends a new line for student 202401237, Lindiwe Zulu, mark 74, to the file. (3)

4.3 Explain the difference between a text file and a database for storing student results. Give ONE advantage of each. (4)

4.4 Explain why exception handling is important in programs used by the public. (2)

4.5 Give TWO examples of situations, other than file errors, where exception handling should be used. (3)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER